Sales Analysis Dashboard Report

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1. Problem Statement

The company is experiencing dynamic growth but facing a decline in sales. The objective is to uncover hidden insights that can support data-driven decision-making and reduce manual effort in data gathering.

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2. AIMS Grid – Project Planning

Purpose: Identify actionable sales insights for decision support and automate reporting to save time.

Stakeholders:

- Sales Director
- Marketing Team
- Customer Service Team
- Data & Analytics Team
- IT (Software Engineering Team)

End Result: An automated dashboard that delivers timely, visualized sales insights for faster, better decisions.

Success Criteria:

- Dashboard reflects real-time sales order insights
- Enable 10% cost savings through better decisions
- Save 20% of analysts' time from manual reporting

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3. Data Engineering Workflow

Tools Used: MySQL, Power BI, Power Query

3.1 Database Connection

- Installed MySQL
- Verified server status
- Imported data using SQL scripts and Power BI's MySQL connector

3.2 SQL Validation Queries

```sql
-- Count total number of transactions
SELECT COUNT(*) FROM transactions;
```

```sql
-- Count total number of customers
SELECT COUNT(*) FROM customers;
```

```sql
-- Filter data for a specific market
SELECT * FROM sales_data
WHERE market_code = 'Mark001';
```

3.3 Data Cleaning & ETL (MySQL + Power Query)

Steps performed:

- Removed irrelevant markets (e.g., New York, Paris)
- Filtered out invalid sales (sales_amount <= 0)
- Currency normalization (USD to INR)
- Removed malformed values (e.g., INR , USD)
- Deduplicated transaction data

Currency Normalization in Power Query (M Code):

```
= Table.AddColumn(#"Filtered Rows", "norm_sales_amount",
    each if [currency] = "USD" or [currency] = "USD#(cr)" then
        [sales_amount]*75
    else [sales_amount])
```

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4. Data Modeling

Schema Type: Star Schema

Fact Table:

- sales_transactions

Dimension Tables:

- sales_customers
- sales_markets
- sales_products
- date

Relationships were established between the fact and dimension tables in Power BI for accurate filtering and aggregation.

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5. DAX Measures

5.1 Base Measures

```
-- Total revenue in INR
Revenue = SUM('sales transactions'[norm_sales_amount])
```

```
-- Total quantity sold
Sales Qty = SUM('sales transactions'[sales_qty])
```

5.2 Profit Calculations

```
-- Total profit
Total Profit Margin = SUM('sales transactions'[profit_margin])
```

```
-- Profit margin percentage
Profit Margin % = DIVIDE([Total Profit Margin], [Revenue], 0)
```

5.3 Contribution Calculations

```
-- Profit margin contribution percentage
Profit Margin Contribution % =
DIVIDE(
    [Total Profit Margin],
    CALCULATE([Total Profit Margin],
        ALL('sales customers'),
        ALL('sales markets'),
        ALL('sales products'))
)
```

```
-- Market revenue share
Market Revenue Share =
DIVIDE(
    [Revenue],
    CALCULATE([Revenue],
        ALL('sales customers'),
        ALL('sales markets'),
        ALL('sales products'))
)
```

5.4 Target Difference Calculation

```
-- Difference from profit target
Target Diff = [Profit Margin %] - 'Profit Target'[Profit Target Value]
```

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6. Dashboard Features

- Year filter (e.g., 2020 slicer)
- Regional filtering and market-wise analysis
- Profit margin tracker by customer and market
- Top 5 customers and products by revenue
- Revenue and profit trends by year/month
- Target simulation via slider
- Bookmarks for preset regional views

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7. Insights & Validation

2020 Key Insights

- **Total Revenue:** ~INR142.2M (validated using SQL)
- **Top Customer:** Electricalsara Stores (46.1% revenue, 50.6% profit)
- **Profit Drop:** Declined from 4% (January) to 0.9% (April)
- **Loss-Making Markets:** Lucknow, Bengaluru
- **Emerging Markets:** Bhubaneshwar (+0.09), Hyderabad (+0.06)
- **Balanced Market:** Mumbai (14.1% revenue, 12.1% profit)
- **Low-Profit Customer:** Electricalssale Stores (0% profit)
- **Client Efficiency Comparison:** Atlas Stores (4.5% margin) vs. Control (3.2%)
- **Top Performing Zones:** South, Central, North (highest profit margins)

SQL for Revenue Validation:

```sql
-- Validates 2020 total revenue across currencies using date join
SELECT SUM(transactions.sales_amount)
FROM transactions
JOIN date ON date.date = transactions.order_date
WHERE year = 2020
AND (currency = 'INR
' OR currency = 'USD
');
```

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8. Outcome

Delivered a validated, dynamic Power BI dashboard that:

- Enables faster, more informed decisions
- Highlights inefficiencies and low-performing segments

- Supports strategic, data-driven growth initiatives